Congenital Heart Diseases & Shock

Yasir Eltahir

CONGENITAL HEART DISEASE (CHD)

Definition

- **Congenital Heart Disease (CHD):**
  Structural or functional heart defect present at birth
- **Etiology:**
  - Genetic factors (chromosomal abnormalities)
  - Environmental factors (maternal illness, medications, infections)

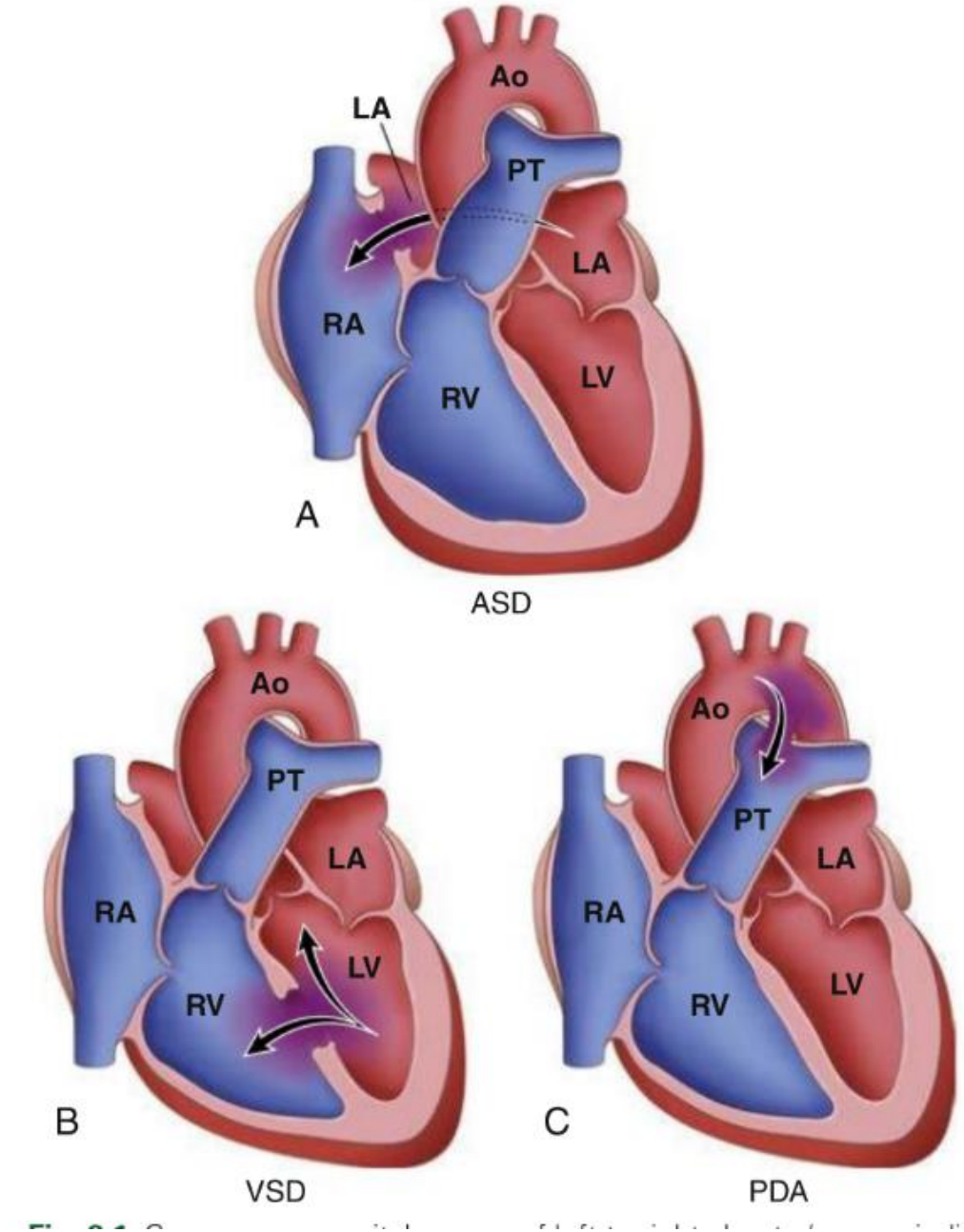
CLASSIFICATION

Based on Hemodynamics:

- **Cyanotic CHD** → Right-to-left shunt → ↓ oxygenation
- **Acyanotic CHD** → Left-to-right shunt → ↑ pulmonary blood flow
- **Obstructive lesions** → Impaired blood flow (no shunt)

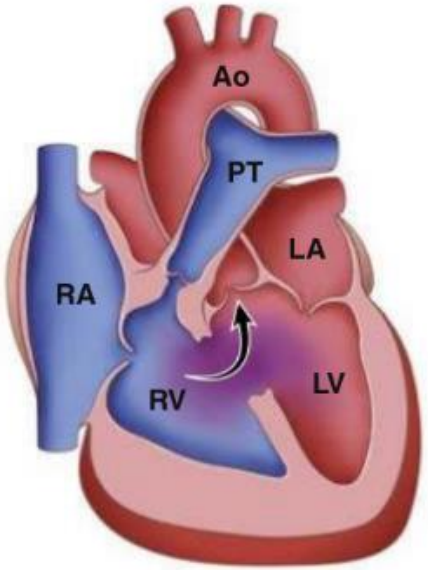
COMMON ACYANOTIC CHDS

Defect	Pathophysiology	Clinical Features
Ventricular Septal Defect (VSD)	Left → right shunt; ↑ pulmonary blood flow	Murmur, HF signs
Atrial Septal Defect (ASD)	Left → right shunt; ↑ RA & RV workload	Fatigue, recurrent infections
Patent Ductus Arteriosus (PDA)	Blood flows aorta → pulmonary artery	Continuous murmur, tachypnea

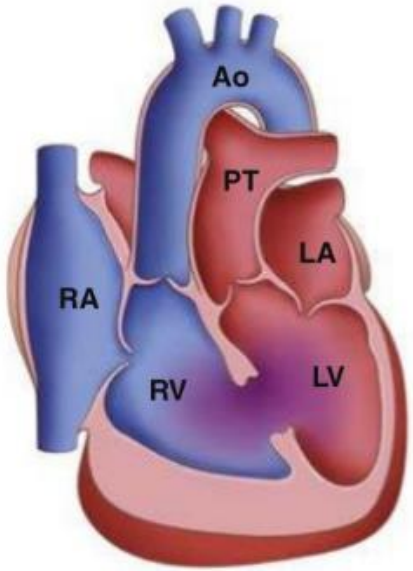

Ao
LA
PT
LA
RA
LV
RV
A
ASD
Ao
PT
LA
RA
LV
RV
B
VSD
Ao
PT
LA
RA
LV
RV
C
PDA

COMMON CYANOTIC CHDS

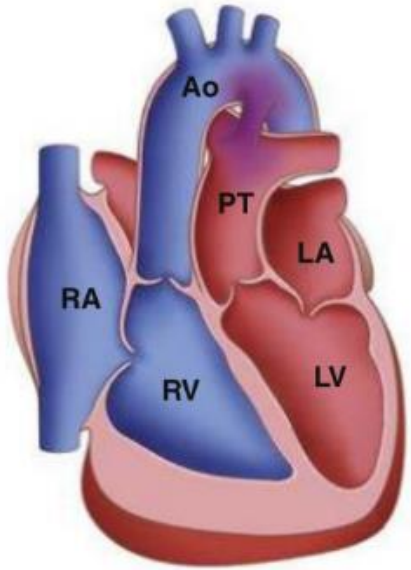
Defect	Pathophysiology	Clinical Features
Tetralogy of Fallot (TOF)	VSD + Pulmonary stenosis + RV hypertrophy + overriding aorta	Cyanosis, "Tet spells"
Transposition of Great Arteries (TGA)	Aorta arises from RV, Pulmonary artery from LV	Cyanosis at birth, HF
Tricuspid Atresia	Absent tricuspid valve → R→L shunt	Cyanosis, growth retardation



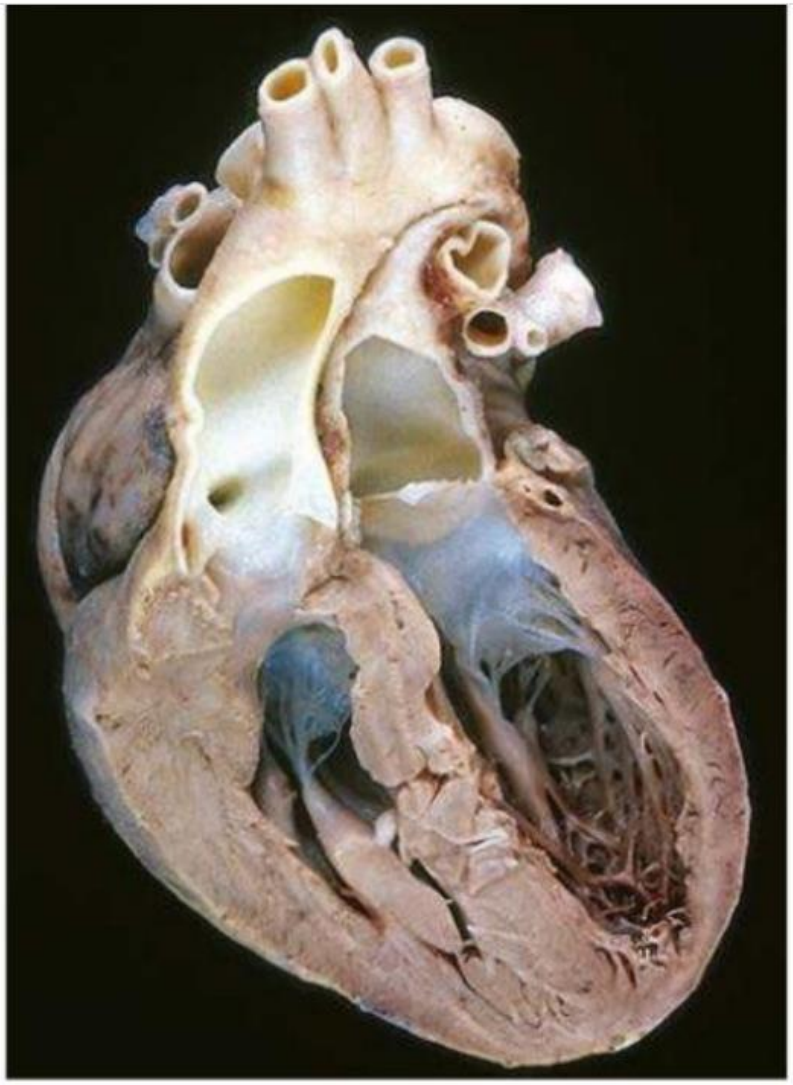
A Classic tetralogy of Fallot



With VSD



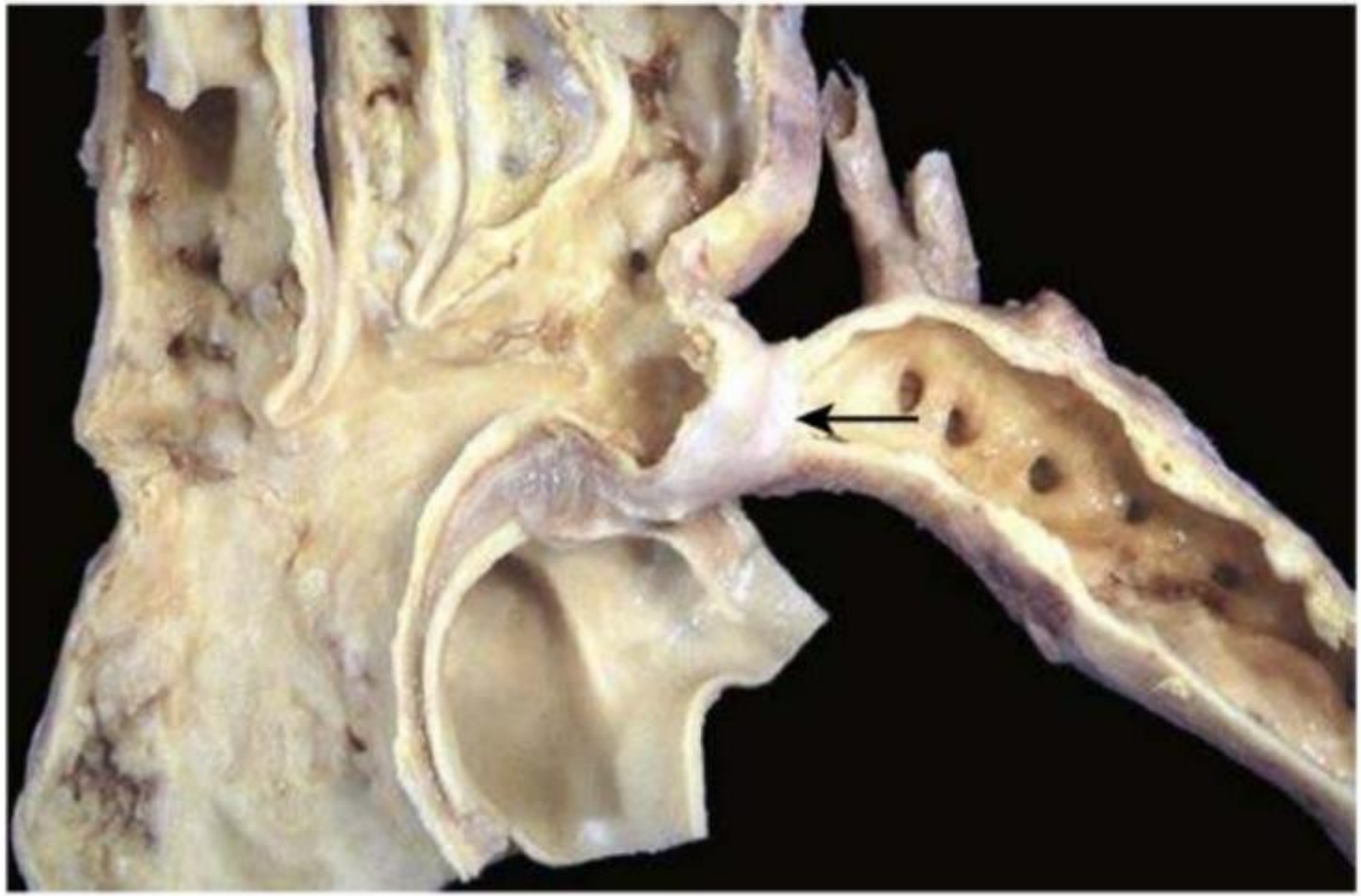
Without VSD



Supplemental eFig. 8.2 Transposition of the great vessels. (Courtesy William D. Edwards, MD, Mayo Clinic,

OBSTRUCTIVE LESIONS

Defect	Pathophysiology	Clinical Features
Coarctation of Aorta	Narrowed aorta → ↑ LV pressure	Hypertension in upper limbs, weak lower pulses
Aortic Stenosis	Obstructed LV outflow	Murmur, exercise intolerance
Pulmonary Stenosis	Obstructed RV outflow	Murmur, cyanosis if severe



CLINICAL MANIFESTATIONS (GENERAL)

- **Acyanotic CHD:** fatigue, tachypnea, poor feeding, HF signs
- **Cyanotic CHD:** central cyanosis, clubbing, "Tet spells," syncope
- **Obstructive lesions:** murmur, BP differences, HF

SHOCK

Definition of Shock

- **Shock:** A life-threatening condition in which **tissue perfusion is inadequate**, leading to **cellular hypoxia and organ dysfunction**.
- **Key Concept:** Oxygen delivery < Oxygen demand → Cellular injury

PATHOPHYSIOLOGY

- **Initial Stage:** Reduced perfusion → ↓ oxygen → anaerobic metabolism → lactic acidosis
- **Compensatory Stage:**
  - Sympathetic activation → ↑ HR, vasoconstriction
  - RAAS activation → fluid retention
- **Progressive Stage:**
  - Cellular dysfunction, capillary leak, organ hypoperfusion
- **Irreversible Stage:** Multiorgan failure → death

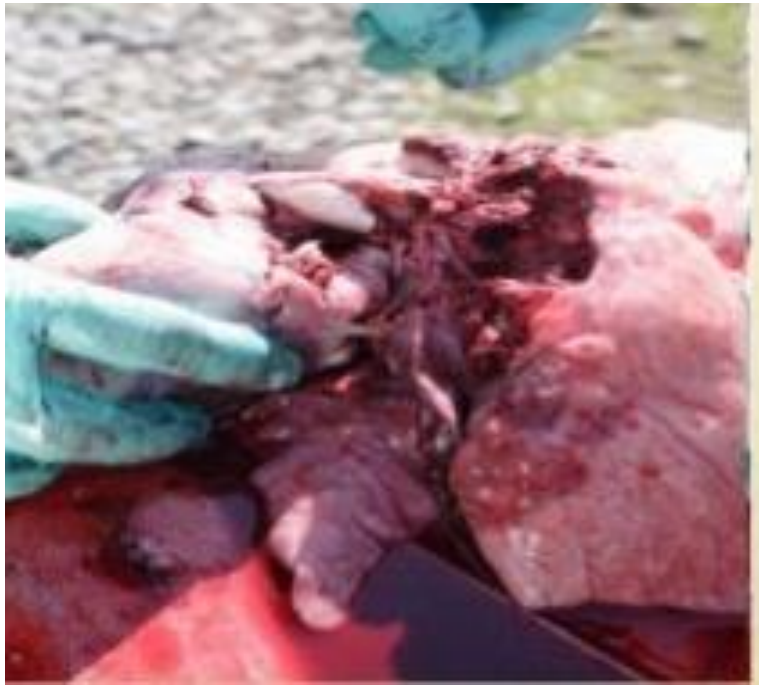
CLASSIFICATION OF SHOCK

Type	Cause	Hemodynamic Effect
Hypovolemic	Blood/fluid loss (hemorrhage, dehydration)	↓ Preload → ↓ CO → ↓ BP
Cardiogenic	Heart pump failure (MI, cardiomyopathy)	↓ CO → ↑ CVP → Pulmonary congestion
Distributive	Abnormal vasodilation (sepsis, anaphylaxis, neurogenic)	↓ SVR → ↓ BP, relative hypovolemia
Obstructive	Mechanical obstruction (PE, cardiac tamponade)	↓ CO → ↑ CVP → ↓ perfusion

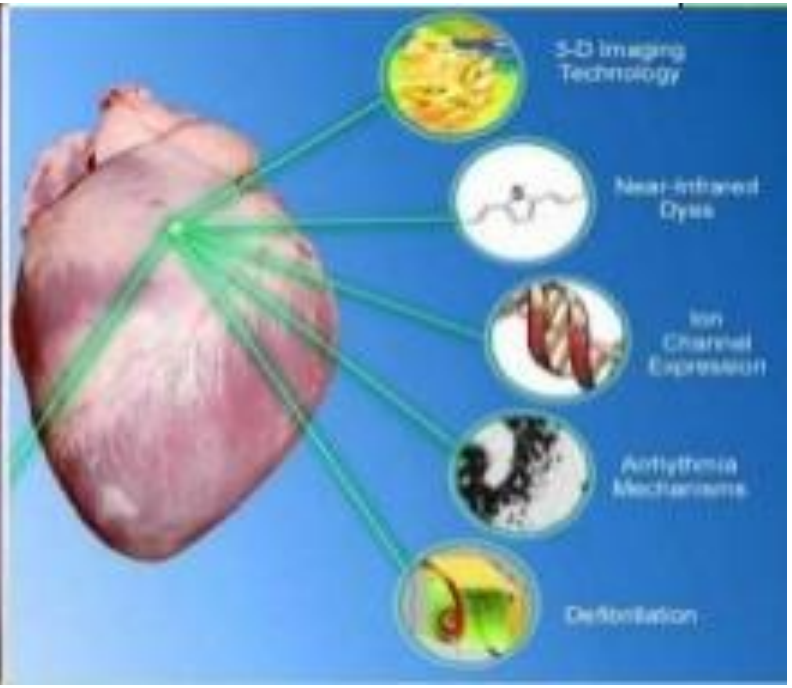
HYPOVOLEMIC SHOCK



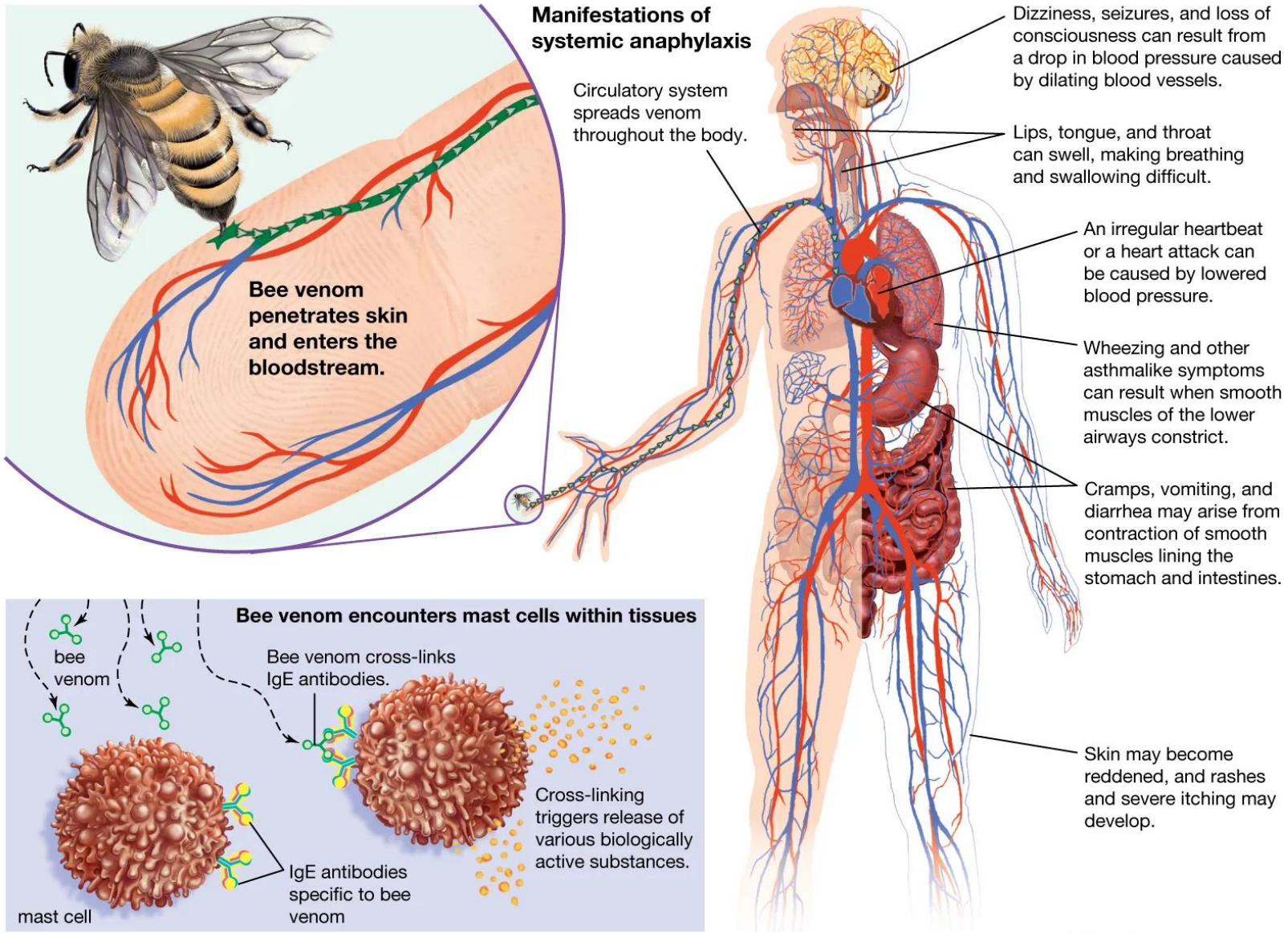
CARDIOGENIC SHOCK





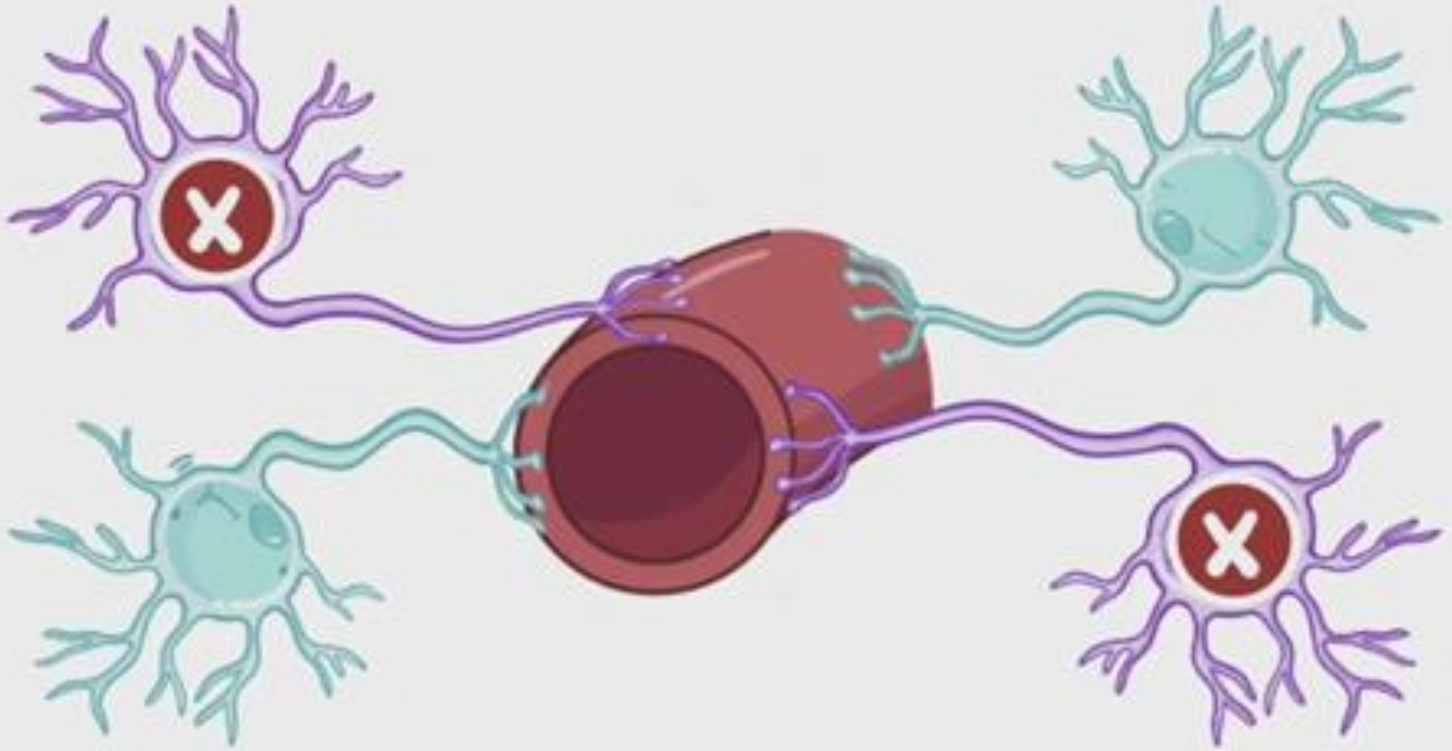


ANAPHYLACTIC SHOCK

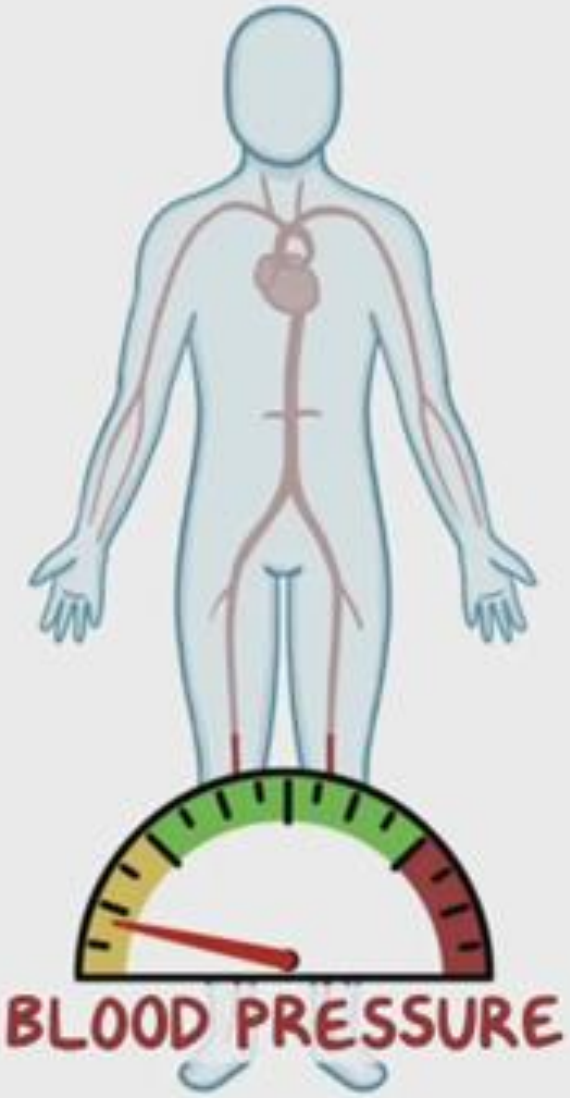


NEUROGENIC SHOCK

LOSS of VASCULAR SYMPATHETIC TONE



UNOPPOSED PARASYMPATHETIC RESPONSE



CLINICAL MANIFESTATIONS & MANAGEMENT

- Tachycardia
- Hypotension
- Cool, clammy skin (except in septic shock)
- Rapid, shallow breathing
- Decreased urine output
- Altered mental status

- Monitor **vital signs, perfusion, urine output**
- Support **airway, breathing, circulation**
- Administer **fluids, medications** as per protocol
- Educate about **early recognition of shock signs**

THANK YOU

